

Data Mining

Clustering: Part 1

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Outline

- ❑ **Overview of Clustering**
- ❑ **Major Clustering Approaches**
 - ❑ **K-means Clustering**
 - ❑ **Hierarchical Clustering**
 - ❑ **DBSCAN Clustering**
- ❑ **Cluster Evaluation**

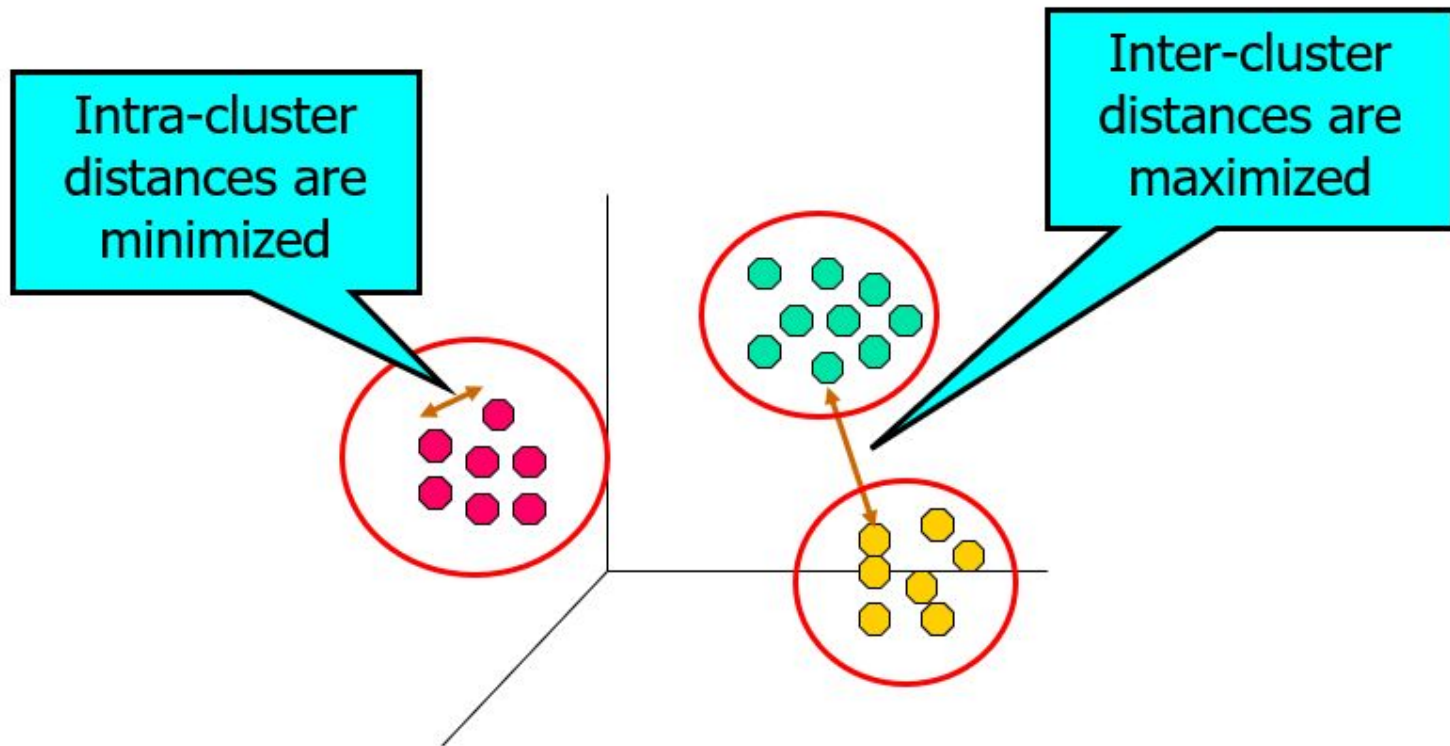
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What is Clustering?

- Given a set of objects, place them in groups such that:

The objects in a group are **similar** (or related) to one another and **different** from (or unrelated to) the objects in other groups



Think of some practical applications of clustering?

Applications of Clustering

- **Marketing:** Discover customer segments for targeted marketing
- **Information retrieval:** Document clustering to enhance search relevance.
- **Land use:** Identifying similar areas in Earth databases based on land use.
- **Biology:** Classifying organisms into taxonomy levels (kingdom to species)
- **City planning:** Grouping houses by type, value, and geographic location.
- **Climate:** Analyzing patterns in atmospheric and oceanic data.
- **Economic science:** Conducting market research through clustering.
- **Etc.**

Clustering as Preprocessing tool

- **Summarization:**
 - Preprocessing for classification, regression, PCA, and association analysis.
- **Compression:**
 - Applied to compress images by clustering similar pixel values or colors.
- **Finding Nearest Neighbors**
 - Localizing search of neighbor objects to one or a small number of clusters.
- **Outlier and Anomaly detection**
 - Outliers are often viewed as data points that are “far away” from any cluster.

What is a **good** clustering and what are the **factors** that contribute to it?

What is a Good Clustering? (1)

- A **good clustering** method will produce high-quality clusters
 - High intra-cluster similarity: **cohesive** within clusters
 - Low inter-cluster similarity: **distinctive** between clusters
- The **quality** of a clustering method depends on:
 - The similarity measure used by the clustering algorithm
 - The implementation of the clustering algorithm
 - The ability to discover some or all of the hidden patterns

What is a Good Clustering? (2)

- **Dissimilarity/Similarity metric**

- Similarity is expressed in terms of a distance function $d(i, j)$
- The definitions of **distance functions** depend on the attribute type: boolean, categorical, interval-scaled, ordinal ratio, and vector variables
- Weights should be associated with different attributes based on the domain application and data semantics

- **Quality of clustering**

- There is a “quality” function that measures the “goodness” of a cluster
- It is hard to define “similar enough” or “good enough” due to subjectivity

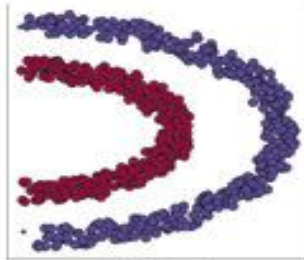
Considerations for Clustering

- **Partitioning criteria**
 - Single-level
 - Hierarchical partitioning (often, multi-level partitioning is desirable)
- **Separation of clusters**
 - Exclusive (e.g., one customer belongs to only one region)
 - Non-exclusive (e.g., one document may belong to more than one category)
- **Similarity measure**
 - Distance-based (e.g., Euclidean, Minkowski, Cosine)
 - Connectivity-based (e.g., Density or Contiguity)
- **Clustering space**
 - Full space (often when low dimensional)
 - Subspaces (often in high-dimensional clustering)

Clustering: Requirements and Challenges

- **Interpretability**
 - Explain and use the different clusters
- **Scalability**
 - Clustering all the data instead of samples
- **Deal with different types of attributes**
 - Numerical, binary, categorical, nominal, ordinal, and a mixture of these
- **Constraint-based clustering**
 - The user can provide constraints to guide the clustering process.
 - Use domain knowledge to set input parameters (metric, number of clusters, ...)
- **Others**
 - Ability to deal with noisy data and outliers
 - Ability to detect clusters of any **shape**

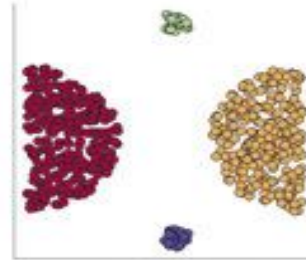
Examples of Cluster Shapes



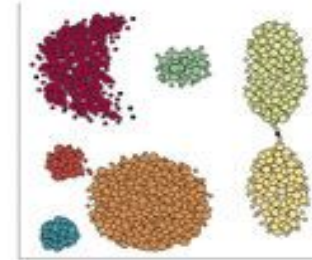
Half-Kernel



Two-Spirals



Outliers



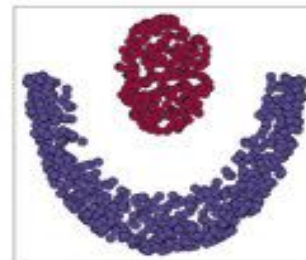
Aggregation



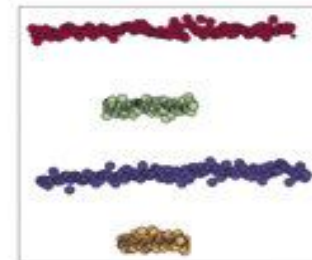
Corners



Cluster-in-cluster



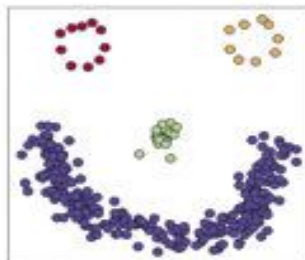
Crescent full moon



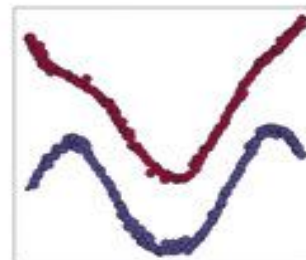
Zelnik5



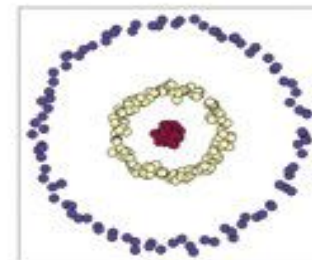
Moon



Face



Wave



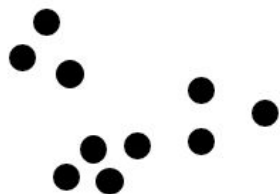
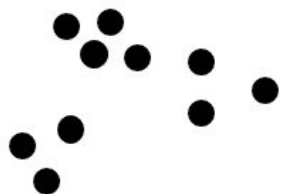
Zelnik1

Notion of a Cluster can be Ambiguous

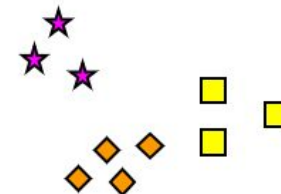
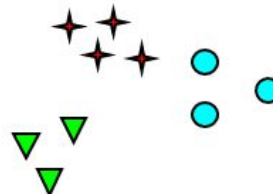


How many clusters?

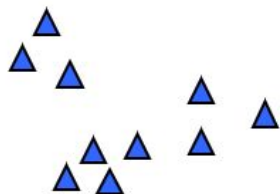
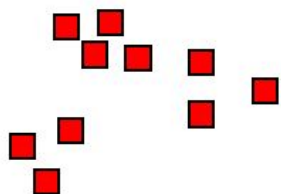
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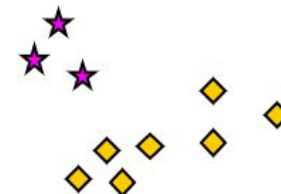
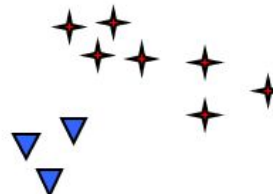
How many clusters?



Six Clusters



Two Clusters

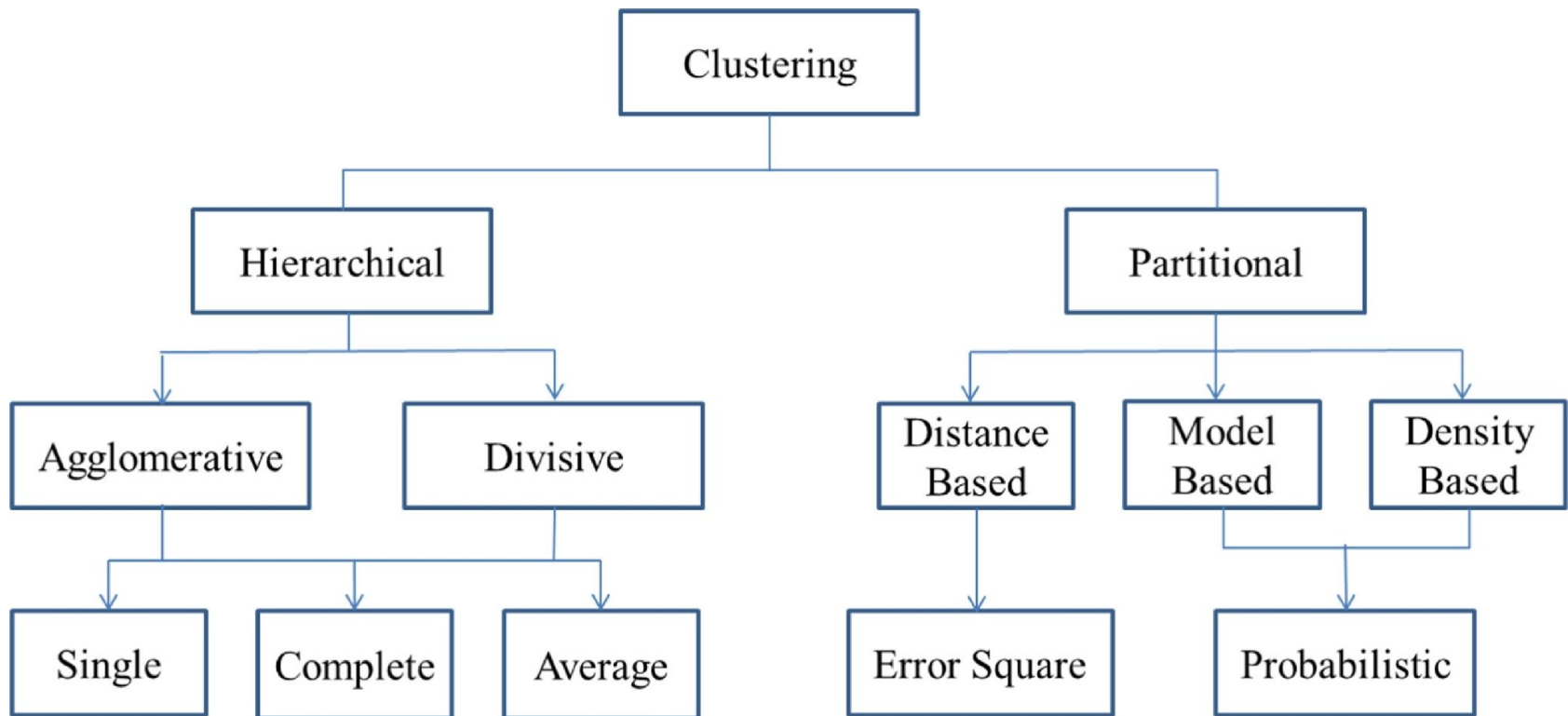


Four Clusters

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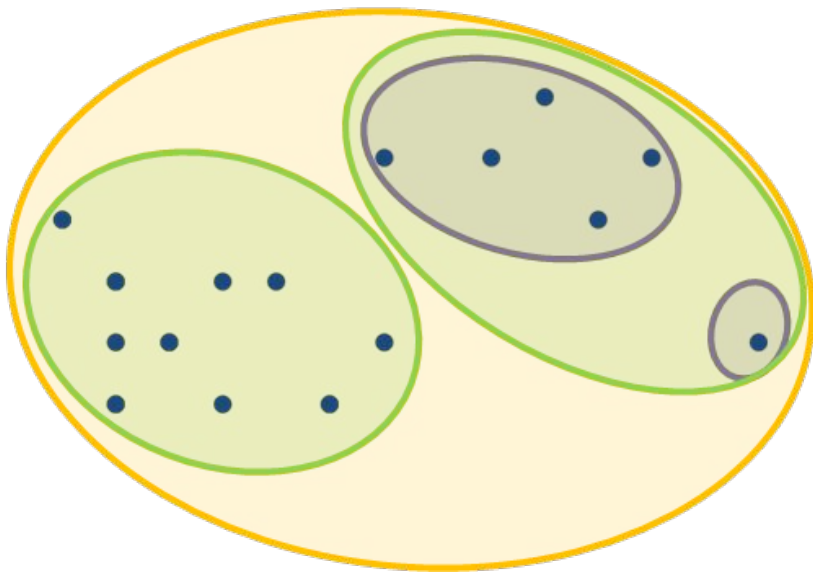
Major Clustering Approaches



In our course, we will focus on Agglomerative, Distance, and Density-based clustering

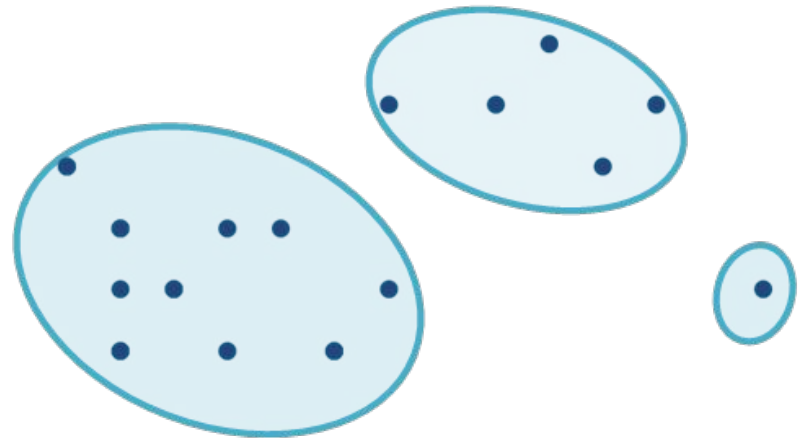
Partitional vs Hierarchical Clustering

Hierarchical Clustering



Nested clusters

Partitional Clustering



Non-nested clusters

Major Clustering Approaches

- **Partitioning approach:**

- Construct various partitions and evaluate them by some criterion
- Typical methods: **K-means**, K-medoids, CLARANS

- **Hierarchical approach:**

- Create a hierarchical decomposition of the set of data using some criterion
- Typical methods: **Diana**, **Agnes**, BIRCH, CAMELEON

- **Density-based approach:**

- Based on connectivity and density functions (detect regions where points are concentrated)
- Typical methods: **DBSCAN**, OPTICS, DenClue

- **Model-based:**

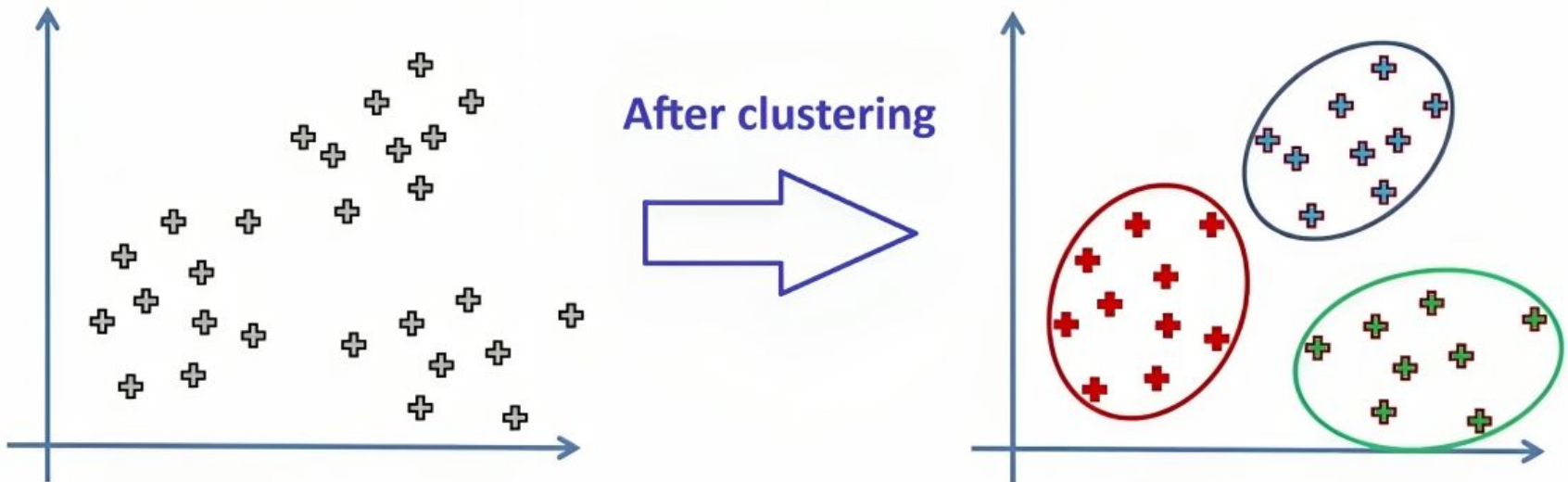
- A model is hypothesized for each of the clusters and tries to find the best fit
- Typical methods: EM, SOM, COBWEB

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Partitional Algorithms

- **Objective:** Partitioning a database D of n objects into a set of K clusters.
- K-Means algorithm is an example of a partitional clustering algorithm.
- Example of clustering data points with $K=3$:



K-Means Algorithm

- The number of clusters K must be specified as input
- Each cluster is represented with a centroid (i.e. center point, e.g. the mean)
- In the first iteration, the K centroids are K random points (objects) from the data
- Each point in the data is assigned to the cluster with the closest centroid
- The centroid of each cluster is updated at each iteration
- The algorithm keeps iterating until the centroid don't change

-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

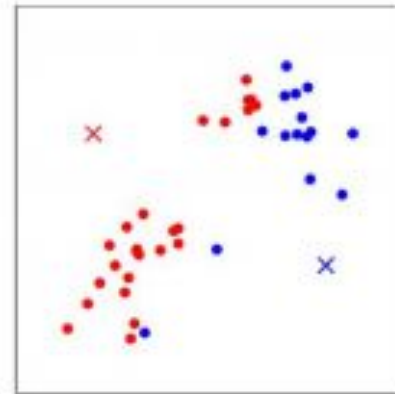
Example of *K*-Means ($K=2$)



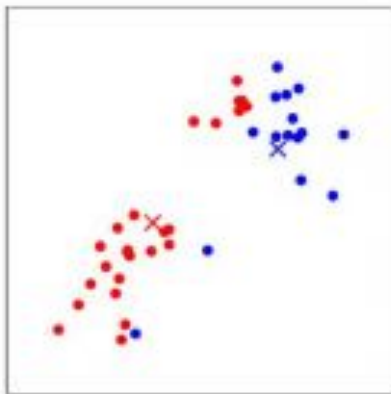
(a)



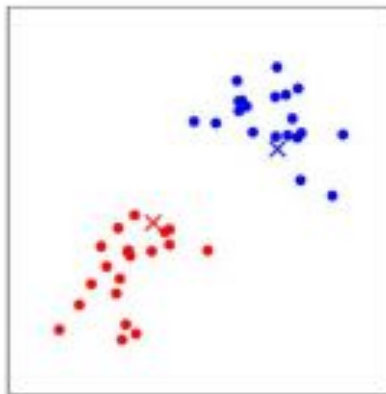
(b)



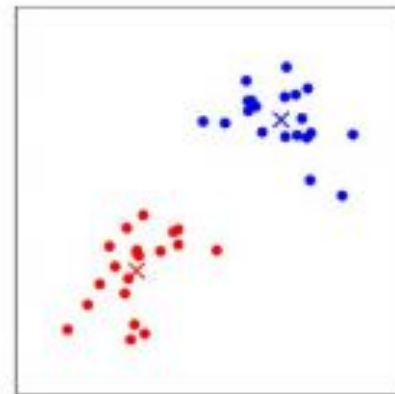
(c)



(d)

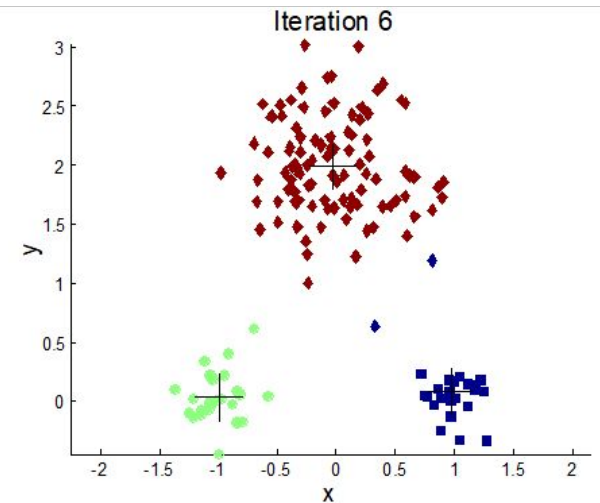
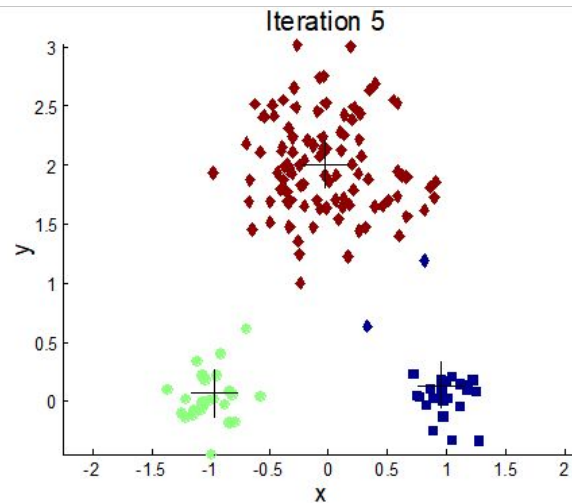
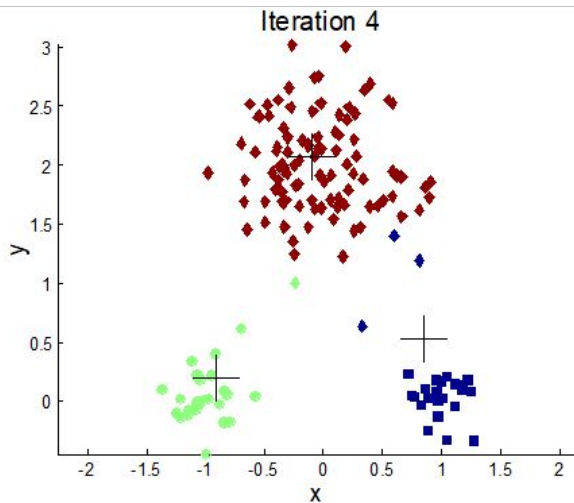
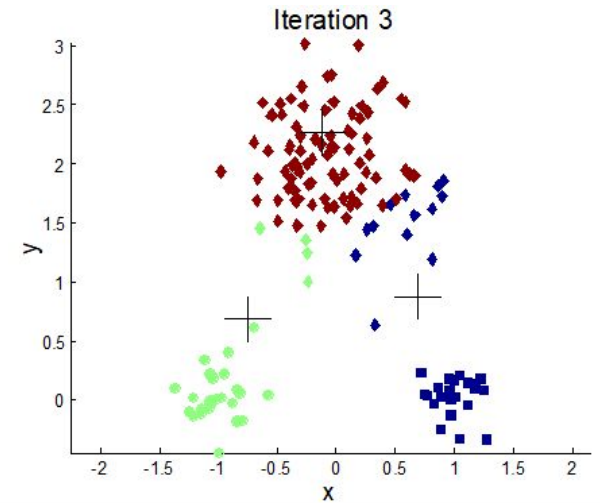
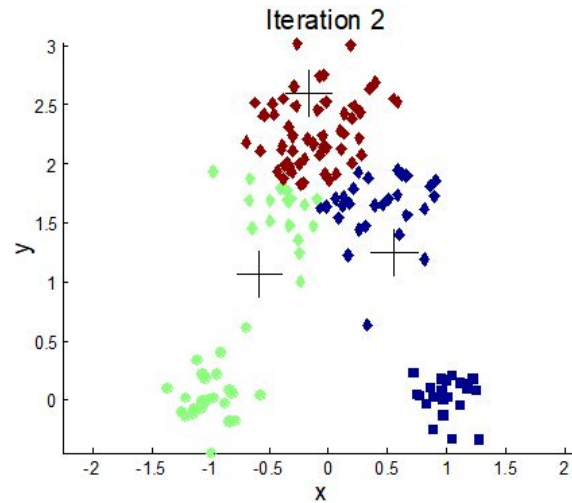
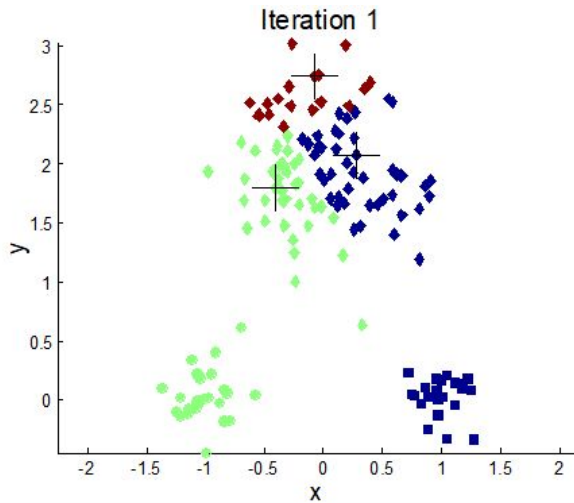


(e)



(f)

Example of K -Means ($K=3$)



Other Examples: K-Means Visualization

This website visualizes the K-Means clustering algorithm:

<https://www.naftaliharris.com/blog/visualizing-k-means-clustering/>

You can explore the website to:

- Select the shape and distribution of the data
- Run K-Means with different values of ***K***
- Set initial centroids and observe the clustering process
- Compare the clustering results of different clustering algorithms

Which objective function should be used?

Objective Function

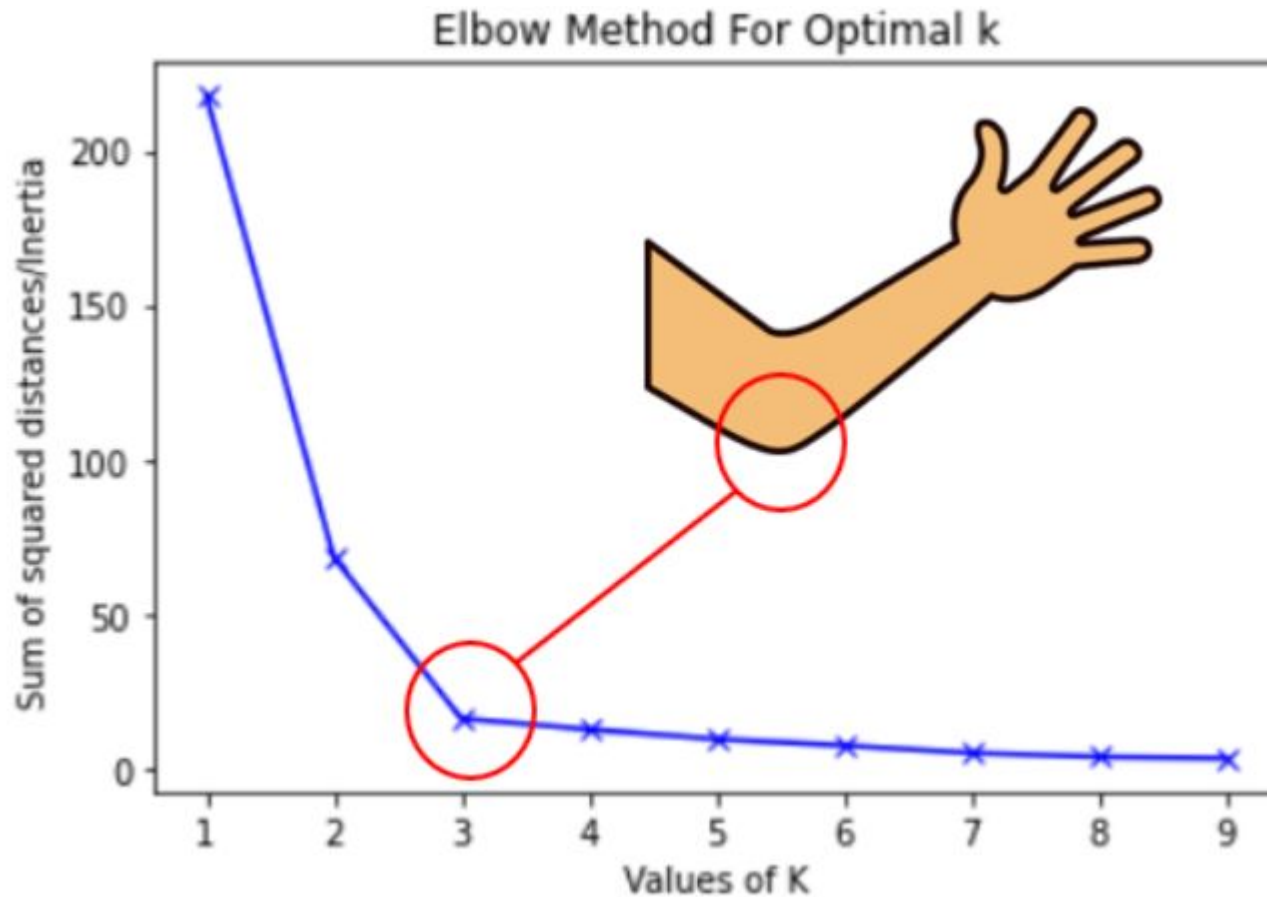
- A common objective function is minimize the Sum of Squared Distances (SSE)
- SSE is used with the Euclidean distance measure

$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist^2(m_i, x)$$

- $\mathbf{x}$ is a data point in cluster $\mathbf{C}_i$ and $\mathbf{m}_i$ is the centroid or medoid for cluster $\mathbf{C}_i$
- Medoid is a real data point, whereas a centroid is not necessarily real
- For each point $\mathbf{x}$, the error is the distance to the nearest cluster center $\mathbf{m}_i$
- To get SSE, we square these errors and sum them
- SSE improves in each iteration until it reaches a local or global minima

How to select the best number of clusters ***K***?

Elbow Method



Line plot between K and inertia

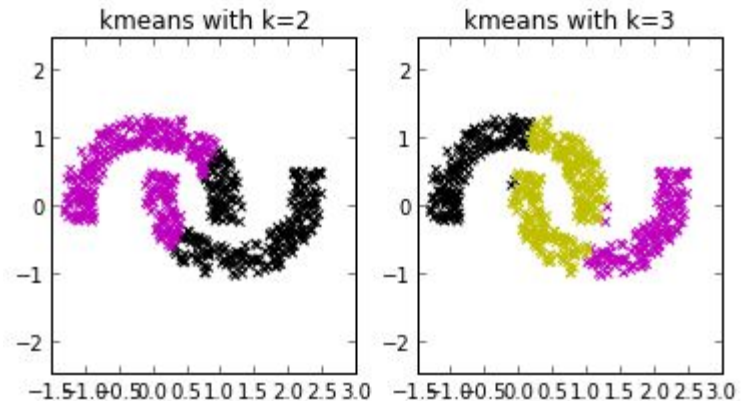
Strength and Weakness of K-Means

- **Strength:** *Fast: $O(tkn)$*

- n : number of objects, k : number of clusters, t : number of iterations
- Normally: $k, t \ll n$

- **Weakness**

- Need to specify k , the *number* of clusters, in advance
- The random choice of the first k centroids may result in different clustering
- Applicable only to objects in a continuous n-dimensional space
- Often terminates at a local optimal
- Sensitive to noisy data and outliers
- Not suitable for non-convex shapes

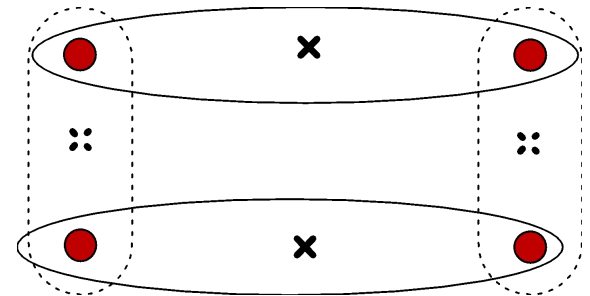


How to improve the K-Means algorithm?

Variations of the *K-Means* Algorithm

- Most of the variants of the *k-means* differ in:

- Selection of the initial ***k*** centroids
- Dissimilarity calculations
- Strategies to calculate cluster means



- Handling categorical data with *k-modes*:

- Replacing means of clusters with modes
- Using new dissimilarity measures to deal with categorical objects
- Using a frequency-based method to update modes of clusters
- A mixture of categorical and numerical data: *k-prototype* method